Lecithin

In subject area: Food Science

Lecithin is defined as a type of phospholipid that serves as a good emulsifier, commonly found in egg yolk, which enhances the stability and quality of food products such as salad dressings.

AI generated definition based on: Introduction to the Chemistry of Food, 2020

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Chapter

Rice Bran Lecithin: Compositional, Nutritional, and Functional Characteristics

Introduction

The word *lecithin* is derived from the Greek term *lekithos*, meaning "egg yolk." In 1846 Gobley isolated lecithin from egg yolk and in 1850 gave it its present name. Lecithin is a natural complex mixture of phosphatides that varies in color from light tan to dark reddish brown and in consistency from a fluid to a plastic solid. The most acceptable definition for lecithin is "a mixture of phospholipids (PLs) extracted from foods of vegetable or animal origin with a minimum of 60% acetone insoluble substances" (Schneider, 2006). Lecithin is the gummy material contained in crude vegetable oils that is removed by degumming. Globally, lecithins are being used as emulsifiers, wetting and instantizing agents, viscosity modifiers, releasing agents, separating agents, and anti-dusting agents. They are also utilized in nutritional supplements and to extend the shelf life of food products. Further, highly purified phospholipids can be used as raw materials for the preparation of lipid vesicles, liposomes, which are the vehicles of choice in some drug delivery systems and are also being explored in the development of artificial lung surfactants (Yukihiro Namba, 1993). Presently soybean, rapeseed, sunflower, corn, and other lecithins are exploited for several applications. However, soybean lecithin is the most commonly used lecithin for various applications. Worldwide soybean production forecasts for 2012–2013 were 271 million metric tons (USDA, May, 2011). Although the lecithin market grows steadily, only an estimated 15–20% of the potential gums are processed into standard quality lecithin and specialties with dedicated technological and nutritional functions in high value pharmaceutical, food, and feed segments (van Nieuwenhuygen, 2014). The widespread planting of genetically modified (GM) soybeans, which began in 1995, has caused disruptions in supply streams of commercial food-grade lecithin from traditional soybean varieties that previously flowed from the United States and Latin America to Europe. As the availability of traditional non-GM soybeans from the United States and Latin America has decreased, the sourcing of identity-preserved soybean lecithin from other regions and lecithin from sunflower and rapeseed has grown (van Nieuwenhuygen, 2014). Hence, rice bran lecithin (RBL) may be an attractive source for those who prefer non-GM-based lecithins.

Even though RBL has high potential for food and industrial applications, this has not enough attention. Therefore, our objective in preparing this chapter is to enhance the knowledge on RBL with respect to its composition of major and minor constituents and its nutritional properties.

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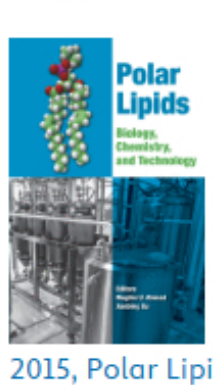

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Ram Chandra Reddy Jala, R.B.N. Prasad

Chapter

Soybean Lecithin: Food, Industrial Uses, and Other Applications

Publisher Summary

Lecithin has long been an important component of a myriad of both food and non-food products and is one of the most versatile and valuable byproducts of the oilseed industry. In foods, lecithin provides about a dozen functions, including as an emulsifier, as a wetting agent, for viscosity reduction, as release agents, and for crystallization control. Lecithin also provides functions in numerous industrial applications as well. By 1940, the U.S. lecithin industry was well established. The lecithin industry is a mature one, but several factors have affected it. Although historically soybean has been the major source of lecithin worldwide, others are being sought because of increased demands for non-Genetically Modified Organism (GMO) lecithin, including canola and sunflower. Although lecithin from GMO soybeans has been shown to be equivalent to non-GMO lines, the European market prefers non-GMO lecithin. Over the past few decades, lecithin has become more important as a neutraceutical and food supplement ingredient. Moreover, the discovery of liposomes has provided a new and more efficient means for drug delivery. This chapter reviews the lecithin industry, manufacture and properties of commercial products, their quality control and modification, and food and nonfood uses.

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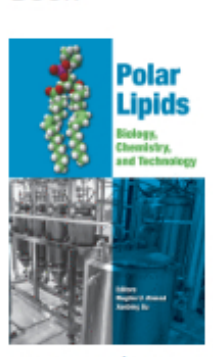

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Chapter

Production and Utilization of Natural Phospholipids

Future Trends

Lecithin is a co-product of the oilseed milling industry. The availability of large volumes of lecithin gums from seed oil degumming and refining challenges further economic valorization of lecithins as sustainable surface-active agents in many applications in food, feed, cosmetics, pharmaceutical, and technical industries.

European sourcing requirements for identity preserved (IP) non-GMO soy lecithin for the European food market drastically changed the lecithin world market supply system beginning in 1996. Traditional non-GMO soybean availability will become more scarce, which presents a market opportunity for high-quality IP non-GMO canola, soy, and sunflower lecithins.

PC fractions with specific essential omega fatty acid profiles, sourced from milk, fish, krill oil, fungi, and algae, are in the research and development pipeline or have reached the market stage for use as nutritional ingredients and health supplements. Extensive clinical studies are required for proving and substantiating health claims, and dossiers have to be approved by the regulatory safety authorities before placing these dedicated lecithins on the market.

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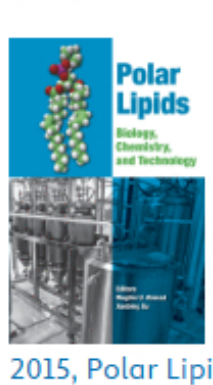

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Chapter

Soybean Lecithin: Food, Industrial Uses, and Other Applications

Introduction

Lecithin has long been an important component of a myriad of both food and nonfood products and is one of the most versatile and valuable byproducts of the oilseed industry. In foods, lecithin provides about a dozen functions, including as an emulsifier, as a wetting agent, for viscosity reduction, as release agents, and for crystallization control. Lecithin also provides functions in numerous industrial applications as well (Hilty, 1948; Sipos, 1989). Much of the lecithin technology was developed in Germany by Bollman and Rewald who held a number of patents describing the process of recovering phosphatides from oilseeds(Bollmann, 1923, 1928, 1929). Bollman also patented a process for improved salad oil manufacture (Bollman, 1926). By 1940, the U.S. lecithin industry was well established (Wendel, 2000). The emulsifying and other functional properties of lecithin were well-recognized and incorporated in numerous foods including margarine, shortenings, baked goods, chocolates, candy, macaroni and noodles, ice cream, instant cocoa beverages, powdered milk, and as crystal inhibitors and antioxidants in fats and oils (Eichberg, 1939; Wiesehahn, 1937). Moreover, a number of nonfood uses were described, including petroleum lubricants, textiles, leather, coatings, rubber, resins, and soaps. The benefits of lecithin were recognized by the cosmetic and pharmaceutical industries as well (Wittcoff, 1951). Phosphatides may undergo numerous chemical reactions including autoxidation, complexing with metal ions, and reactions within the fatty acids or the phosphatide groups (Ghyczy, 1989; Pryde, 1985). However, only a few have been exploited by the industry.

Much of the work on chemical modification of phosphatides, including sulfonation (Scharf, 1953), halogenation (Denham, 1939), hydroxylation (Wittcoff, 1948, 1949), hydrogenation (Arveson, 1942), acetylation (Davis, 1970, 1971; Hayes and Wolff, 1957), alcohol fractionated (Julian and Iveson, 1958), acid/base hydrolyzed (Davis, 1970, 1971), blending (Wiesehahn, 1940), and oil free lecithin (Thurman, 1947) have been detailed in the patent literature.

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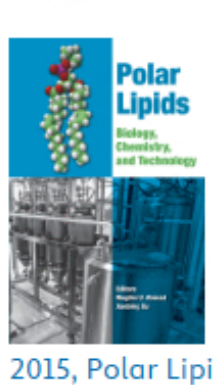

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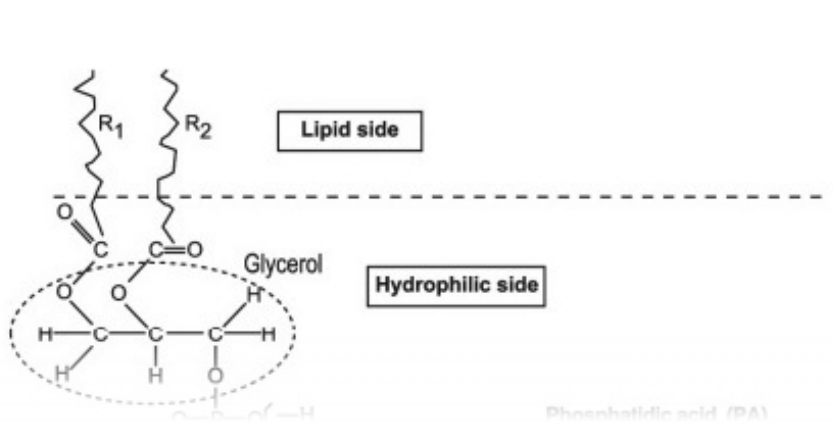
Effect of Emulsifiers on Cocoa Butter and Chocolate Rheology, Polymorphism, and Bloom

Lecithins as a Rheology Modifier

Lecithin was discovered in 1845–1874 by Théodore Nicolas Gobley (Gobley, 1845a, 1845b, 1847, 1850, 1874a, 1874b, 1874c), who isolated it in egg yolks and identified it as the substance that allowed oil and water to mix. The name is derived from the Greek word *lekithos*, which means "yolk of egg." Lecithin is a naturally occurring fatty substance found in several foods including soybeans, whole grains, and egg yolks. It is often used as an emulsification agent in processed foods. It can be taken in various forms as a nutritional supplement, often derived from soybeans. The body breaks lecithin down into its component parts: choline, phosphate, glycerol, and fatty acids. The body's greatest concentration of lecithin is found in the vital organs, where it makes up about 30% of the dry weight of the brain. Lecithin acts as an emulsifier and helps the body in the absorption of fats.

The lecithin that is used in chocolate is a commercial product containing ca. 50–65 wt% phospholipids (PL); the rest is oil with some minor components. The PL is composed mainly from phosphatidylcholine (PC, ca. 21%) along with other PL derivatives such as phosphatidylethanolamine (PE, 22%), phosphatidylinositol (PI, 19 wt%), phosphatidic acid (PA, ca. 6%), phosphatidylglycerol (PG), phosphatidylserine (PS), and some other minor components (Fig. 12.6).



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2012, Cocoa Butter and Related Compounds

Nissim Garti, Abraham Aserin

Chapter

Emulsifiers as food texture modifiers

2.2.1.5 Lecithin

In contrast to these synthetic emulsifiers, lecithin is a natural food emulsifier that can be extracted from a variety of sources, including soybeans, rapeseed and egg. Soy lecithin is the most widely used surfactant ingredient in the food industry because it can be economically extracted during the processing of crude soybean oil. Natural lecithins contain a complex mixture of different types of phospholipids and other lipids, although they can be fractionated to form purer ingredients that are enriched with particular fractions. The most common phospholipids in lecithin are PC, phosphatidylethanolamine (PE) and phosphatidylinositol (PI). The hydrophilic head groups of these molecules are either anionic (PI) or zwitterionic (PC and PE), whereas the lipophilic tail groups consist of two fatty acids. Natural lecithin has intermediate solubility characteristics and HLB numbers (~8), which means that it is not particularly suitable for stabilising either O/W or W/O emulsions when used in isolation, but it may be effective when used in combination with other surfactants (van Nieuwenhuyzen and Szuhaj, 1998). In addition, lecithin can be chemically or enzymatically hydrolysed to break off one of the hydrocarbon tails to produce more hydrophilic surfactants that are capable of stabilising O/W emulsions.

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2015, Modifying Food Texture

L. Chen

Chapter

PHOSPHOLIPIDS | Properties and Occurrence

Occurrence

The International Lecithin and Phospholipid Society defines lecithin as 'a complex mixture of glycerophospholipids obtained from animal, vegetable or microbial sources, containing varying amounts of substances such as triglycerides, fatty acids, glycolipids, sterols, and sphingophospholipids.' Lecithins are natural surfactants primarily derived from soya beans and eggs. They are found most abundantly in seeds and nuts, eggs, brains, and cell walls, in a concentration range of 0.5–2%. (*See* EGGS | Structure and Composition; FATTY ACIDS | Properties; SOY (SOYA) BEANS | Properties and Analysis; TRIGLYCERIDES | Structures and Properties.)

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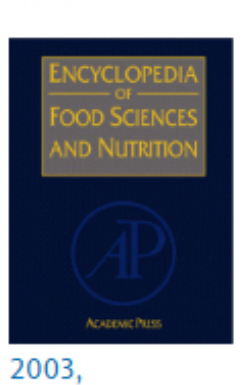

2003, Encyclopedia of Food Sciences and Nutrition (Second Edition)

B.F. Szuhaj

Chapter

Production and Utilization of Natural Phospholipids

Regulatory Specification

According to most legislative definitions for use as food ingredients, lecithins are mixtures of phosphatides (phospholipids) derived from vegetable and animal origins. In the United States, lecithins are classified in the Food Chemical Codex (FCC III). Under Title 21, part 184 of the Code of Federal Regulations, the FDA has regulated the use of lecithin as generally recognized as safe (GRAS) food substances. Standard lecithin, fractions, and acetylated lecithin are listed in §184.1400, enzyme-modified lecithin in §184.1063, and hydroxylated lecithin in §172.814. Table 9.E lists the various purity specifications of food-grade lecithin.

Table 9.E. Legal Purity Specifications of Food-Grade Lecithin

Purity	FAO/WHO Codex Alimentarius	European Union E322	Food Chemical Codex
Acetone insoluble %	>60	>60	>50
Hexane insoluble %		Hydrolyzed > 56	<0.3
Toluene insoluble %	<0.3	<0.3	<0.3
Moisture %	–	–	<1.5
Drying loss %	<2.0	<2.0	<1.5
Acid value mgKOH/g	<36	<35	<36
Peroxide value meq/Kg	<10	Hydrolyzed < 45	<100
Arsenic ppm	<3	<3	<10
Lead ppm	<10	<5	<1
Mercury ppm	–	<1	–
Heavy metals (as Pb) ppm	<40	<10	–

The EU-approved food additive number E322 covers all standard, physical fractionated, and enzymatically hydrolyzed lecithins for use in the food industry. Hydroxylated and acetylated lecithins do not have an EU-approved food additive status.

EU legal specifications are similar or close to the Codex Alimentarius recommended specifications and methods of analysis. The Codex numbers are recommendations to nations for worldwide use; however, national laws are binding.

Regulations for use in health foods, supplements, or pharmaceutical products may require further specific purity specifications.

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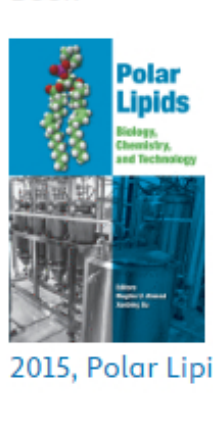

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Chapter

Soybean Lecithin: Food, Industrial Uses, and Other Applications

Beverages

The use of lecithin in beverages has been reviewed by Sander (1989). Lecithinization of dry powders can be done in several ways. Lecithin is a natural surfactant that can be applied on individual ingredients or in a complex formula that may consist of proteins, cocoa powders, trigylcerides, and carbohydrates. Direct incorporation of lecithin can be accomplished by high shear mixing, coblending with spraying onto a powder, or cospraying a total emulsion of ingredients. Agglomeration may be done by continuous batch or fluid bed systems. A formula for a reduced-lactose drink consists of 2% lecithin whey protein concentrate, fructose, maltodextrin, vitamins, and flavors.

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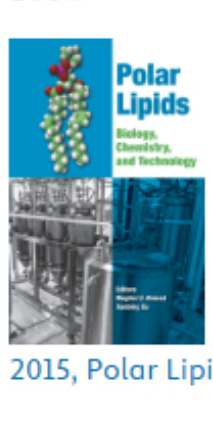

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Chapter

Phosphatidylserine: Biology, Technologies, and Applications

Lecithin

Lecithin, the raw material for manufacturing PS, is easier to obtain from plants than from animal tissues. The most common ones are from soy, rapeseed, and sunflower. Other sources, such as rice, peanut, corn, and palm, are also available. Egg lecithin is the most common one from animal origin. In recent years, marine lecithin has received much attention because of its high content of n-3 fatty acids, especially lecithin from krill. Krill oil has a high content of phospholipids, mainly phosphatidylcholine. Table 6.F lists the composition of the major lecithins on the market.

Table 6.F. Compositions of Major Commercial Lecithins

Empty Cell	Empty Cell	Empty Cell	Rapeseed Oil[d]		Empty Cell	Empty Cell	Empty Cell	Empty Cell
Composition (%)[a]	**Soy[b]**	**Sunflower[c]**	**Hybridol**	**Pactol**	**Egg Yolk[e]**	**Milk[f]**	**Marine[g]**	**Squid[h]**
PC	23	65	6	19	83.2	26	68	80.5
LPC	–	–	–	–	0.7	–	–	–
PE	20	10	38	2	14.2	31.5	23	13.2
PA	8	6	1	25	–	–	–	–
PI	14	3	55	24	–	4.9	2	–
PS	–	–	–	–	–	8.8	2	–
SM	–	–	–	–	–	–	1	–

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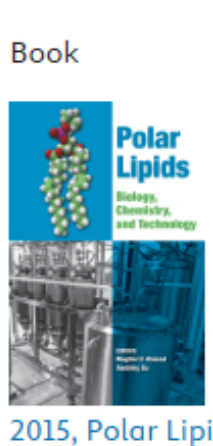

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